Encoding of small-scale models in the medial prefrontal cortex

Maksym Manakov*[1], Mikhail Proskurin*[1], Hanqing Wang[1], Elena Kuleshova[1], Andy Lustig[1], Reza Behnam[1], Shaul Druckmann[2], Gowan Tervo[#1], and Alla Karpova[#1]

[1] Janelia Research Campus; [2] Stanford University; [#] correspondence

problem

The importance of small-scale models of the world and of one's actions in it for intelligent behavior as proposed by Kenneth Craick is well-appreciated, but we still know little about how such models and built and updated by the brain. The likely infinite number of possible world models, the ad hoc nature of the animals' approach to their construction and updating, and the model's necessarily tailoring to the specific needs of the individual make probing the structure of the small-scale model challenging. A particularly useful solution to this problem would come in the form of a behavioral farmework that permits 1) the assessment of the overall model structure, as well as 2) the interpretation of each step in the context of that structure.

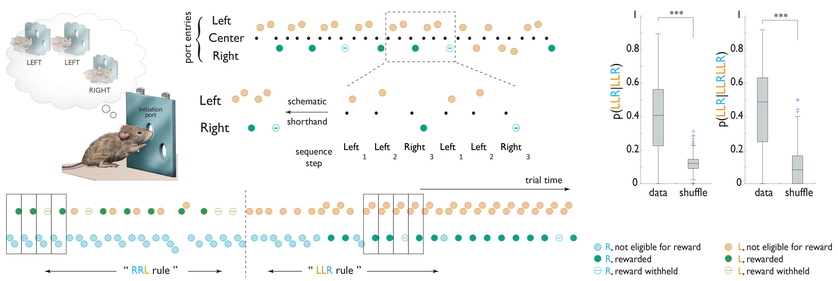
approach

Our approach rested on the intuition that despite organisms may not be as capable at building abstract world models, if a behavioral task could be designed to force reliance on abstract small-scale models, then combining that behavior with monitoring neural activity might uncover a relationship between the structure of neural activity and that model so crystalline as to permit the interpretation at the single step level. Our behavioral task requires rats to acquire, through self-guided exploration in the absence of any external instruction, the concept of a "left" and a "right" trial, and to discover a number of latent target sequences that are preferentially rewarded. During various stages of this dynamic task, we record population activity in the medial prefrontal cortex.

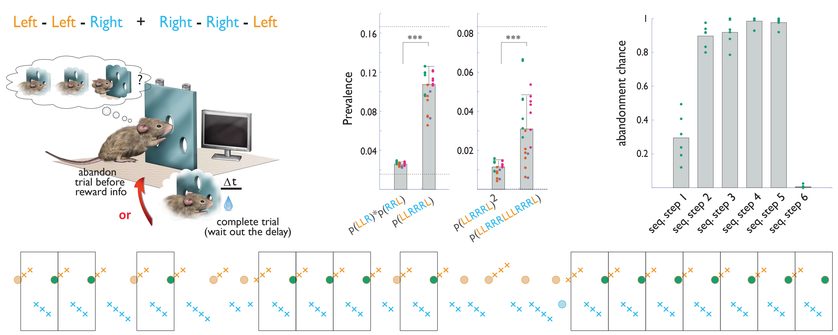
findings /ongoing work

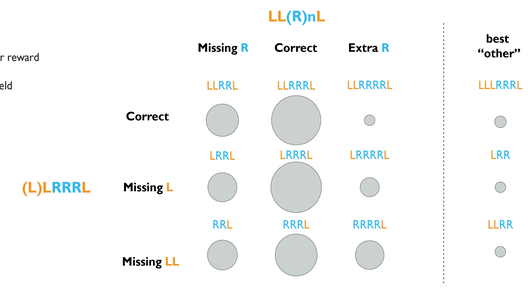
Rats solve this task at a cognitive level, using sequences of "Lefts" and "Rights" as behavioral units with relational structure. Under these conditions, we find a most striking structure in neural activity in mPFC that both constitutes the simplest comprehensive small-scale model and permits behavioral interpretation at the single trial level. Having convinced ourselves that the structure in neural space behaves as we expect it to when the animal's world changes, we are now beginning to use the structure of neural activity as a proxy for the structure of the animal's knowledge in more complex settings and during learning.

self-guided discovery of "left", "right", and sequences

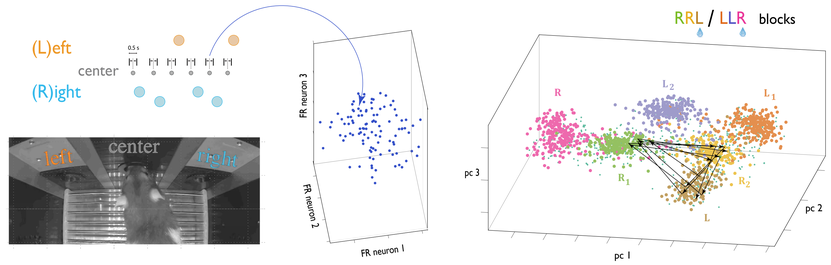


sequences as units with relational structure

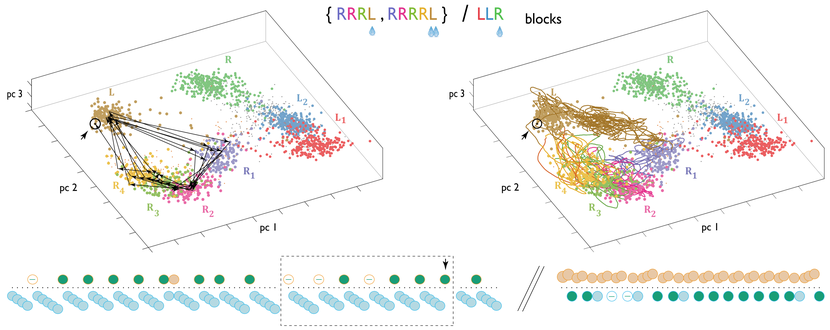




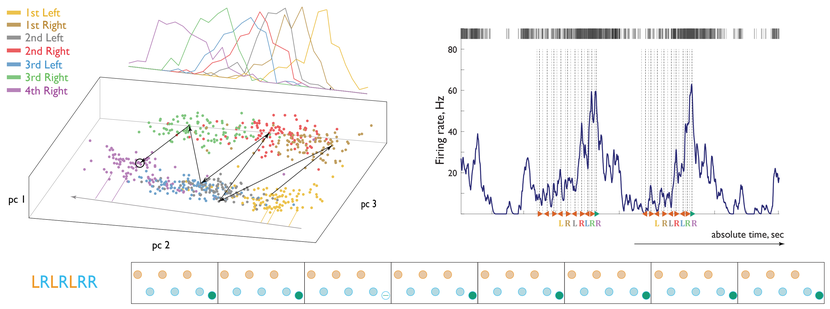
minimal comprehensive representation in mPFC



representation scales with task complexity

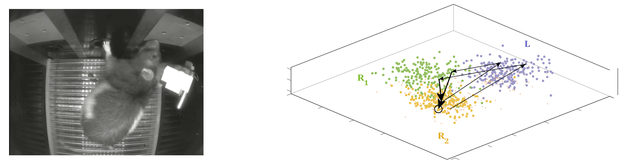


sequence progression dimension

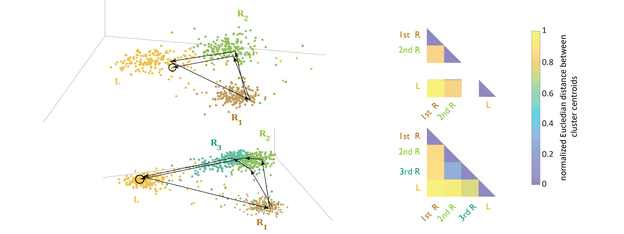


model, not behavioral record

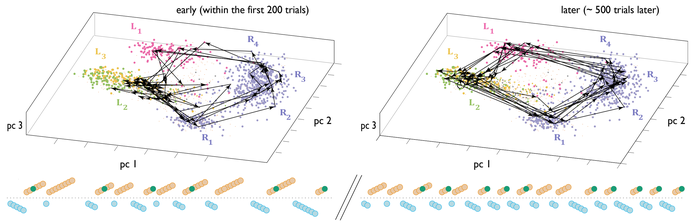
effector independence of cluster progression



cluster geometry after learning (RRRL after RRL)



cluster rearrangement during learning (acquisition of RRRLLL after RRRL)



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